2.5

1

(3) $3 \cdot \frac{dy}{dx} + 12y = 4$

$\frac{dy}{dx} + 4y = \frac{4}{3}$

$F(x) : e^{\int 4\,dx} = e^{4x}$

$e^{4x} \cdot \frac{dy}{dx} + 4 \cdot e^{4x} \cdot y = e^{4x} \cdot \frac{4}{3}$

$\frac{d}{dx}(e^{4x} \cdot y) = \frac{4}{3} e^{4x}$

$e^{4x} \cdot y = \int \frac{4}{3} e^{4x}$

$= \frac{1}{3} e^{4x} + C$

$\therefore y = \frac{1}{3} + c \cdot e^{-4x}$ //

2

(5)

$y' + (\tan x) y = \sin 2x, \; y(0) = 3$

$F(x) = e^{\int \tan x\,dx}$

$\int \tan x\,dx = \int \frac{s}{c}\,dx$

$\cos x = t$

$\int \frac{-1}{t}\,dt = -\ln|t| = -\ln|\cos x|$

$\therefore F(x) = e^{-\ln|\cos x|} = \frac{1}{\cos x}$ //

$\frac{y'}{\cos x} + \frac{\sin x}{\cos^2 x} \cdot (y)$

$= \frac{\sin(2x)}{\cos(x)}$

$\frac{d}{dx}\left(\frac{y}{\cos x}\right) = \frac{2\sin x \cos x}{\cos x}$

$\int \frac{\sin x}{\cos^2 x}\,dx$

$= \int \frac{-1}{t^2}\,dt$

$= \frac{1}{t}$

$\frac{-y \cdot (-\sin x)}{\cos^2 x}$

$\therefore \frac{y}{\cos x} = \int 2\sin x\,dx$

$= -2\cos x + C$

$\therefore y = -2\cos^2 x + C\cos x$ //

$3 = -2 + C$

$C = 5$

$\therefore y = -2\cos^2 x + 5\cos x$ //

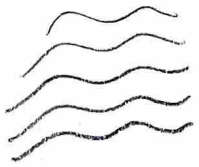
#3

(1) $x\,dy + y^2\,dx = 0$, $y(1) = 0.2$

$y^2\,dx + x\,dy = 0$

$P_y = 2y$

$Q_x = 1$



$\frac{1}{P}(Q_x - P_y) = \frac{1}{y^2}(1-2y)$

$\therefore F(y) = e^{\int \frac{1}{y^2}(1-2y)\,dy}$

$= e^{+\frac{1}{y} + 2\ln y}$

$x \cdot e^{\frac{1}{y}+2\ln y}\,dy + y^2 \cdot e^{\frac{1}{y}+2\ln y}\,dx = 0$

$\frac{1}{Q}(P_y - Q_x)$

$= \frac{2y}{x} - \frac{1}{x}$

$\therefore F(x) = e^{\int \frac{2y-1}{x}\,dx}$

$= e^{2y\ln x - \ln x}$

$= e^{(2y-1)\ln x}$

$= x^{2y-1}$

$\therefore x^{2y-1} \cdot y^2\,dx + x^{2y}\,dy = 0$

$\therefore u(x,y) = \int x^{2y}\,dy + g(x)$

$= \frac{1}{2}x^{2y} + g(x)$

$u_y = x^{2y} + g'(x)$

$x\,dy = -y^2\,dx$

$\int -\frac{dy}{y^2} = \int \frac{1}{x}\,dx$

$+\frac{1}{y} = \ln x + C$ ← $y(1)=2$

$\therefore C = \frac{1}{2}$

$\therefore \frac{1}{y} = \ln x + \frac{1}{2}$

$u(x,y) = \int y^2 e^{\frac{1}{y}+2\ln y}\,dx + g(y)$

$= xy^2 e^{\frac{1}{y}+2\ln y} + g(y)$

$u_y = x\left(2y \cdot e^{\frac{1}{y}+2\ln y} + y^2 \cdot \left(-\frac{1}{y^2}+\frac{2}{y}\right) e^{\frac{1}{y}+2\ln y}\right) + g'(y)$

$(-1+2y)$

$= x \cdot e^{\frac{1}{y}+2\ln y}$

$= x(4y-1)\,e^{\frac{1}{y}+2\ln y} + g'(y)$

$g'(y) = (-4xy+2x)\,e^{\frac{1}{y}+2\ln y}$

$= -2x(2y-1)\,e^{\frac{1}{y}+2\ln y}$

(X)

#3

(1) $xy^2 \frac{dy}{dx} = y^3 - x^3$, $y(1) = 2$ //

$\frac{y^2}{x^2} \cdot \frac{dy}{dx} = \frac{y^3}{x^3} - 1$

$\frac{y}{x} = u \qquad y = ux \qquad \frac{dy}{dx} = \frac{du}{dx} \cdot x + u$

$u^2 \cdot \frac{dy}{dx} = u^3 - 1$

$u^2 \left(\frac{du}{dx} \cdot x + u \right) = u^3 - 1$

$xu^2 \cdot \frac{du}{dx} = -1$

$\int u^2 du = \int -\frac{1}{x} dx$

$\frac{1}{3}u^3 = -\ln|x| + C$

$c \cdot e^{-\frac{1}{3}u^3} = x$

$\therefore c \cdot e^{-\frac{1}{3} \cdot \left(\frac{y}{x}\right)^3} = x$ ///

$c \, e^{-\frac{1}{3} \cdot 8} = 1$

$c = e^{\frac{8}{3}}$

$\therefore e^{\frac{8}{3}} \cdot e^{-\frac{1}{3} \cdot \left(\frac{y}{x}\right)^3} = x$ //

2.4

2

(3) $y^3 dx + 3xy^2 dy = 0$

$M_y = 3y^2$

$N_x = 3y^2$

$\therefore M_y = N_x$

$u(x,y)$

$= \int y^3 dx + g(y)$

$= xy^3 + g(y)$

$u_y = 3xy^2 + g'(y)$

$= 3xy^2$

$\therefore g'(y) = 0$

$g(y) = C$ //

$\therefore xy^3 = C$

2.2

01

(1) $\frac{y^2}{x} \cdot \frac{dy}{dx} = 1+x^2$

$y^2 dy = x(1+x^2)\,dx$

$\int y^2 dy = \int x(1+x^2)dx$

$\frac{1}{3}y^3 = \frac{1}{4}x^4 + \frac{1}{2}x^2 + C$ //

02

(5) $\frac{dy}{dx} = \frac{x-1}{y+2}$, $y(-1) = 6$

$(y+2)\,dy = (x-1)dx$

$\int (y+2)\,dy = \int (x-1)\,dx$

$\frac{1}{2}y^2 + 2y = \frac{1}{2}x^2 - x + C$ //

$y^2 + 4y = x^2 - 2x + C$

$(y+2)^2 = (x-1)^2 + C$ //

$64 = 1 + C$

$\therefore C = 60$

$(y+2)^2 = (x-1)^2 + 60$ //

2.3

2

(3)

$x\,dx + (y-2x)\,dy = 0$

$1\,dx + \left(\frac{y}{x} - 2\right)dy = 0$

$u = \frac{y}{x} \rightarrow y = ux$

$dx + (u-2)\,dy = 0$

$dy = du\,x + u\,dx$

$dx + (u-2)(du\,x + u\,dx) = 0$

$(u^2 - 2u + 1)\,dx + (u-2)\,du = 0$

$\frac{1}{x}dx + \frac{u-2}{(u-1)^2}du = 0$

$-\ln|x|$

$= \int -\frac{u-2}{(u-1)^2}\,du$

$= \int \frac{u-1-1}{(u-1)^2}\,du$

$= \ln|u-1| - \int \frac{1}{(u-1)^2}\,du$

$= \ln|u-1| + \frac{1}{(u-1)} + C$

$\therefore \ln\frac{1}{x}$

$= \ln\left(\frac{y}{x} - 1\right) + \frac{1}{\left(\frac{y}{x} - 1\right)} + C$ //

1.3 연습문제 //

04. 반지름 : r

부피 : $V = \frac{4}{3}\pi r^3$

표면적 : $S = 4\pi r^2$

증발 속도가 표면적에 비례하므로,

$$\frac{dV}{dt} = -kS \;//$$

$$\frac{dV}{dt} = \frac{d}{dt}\left(\frac{4}{3}\pi r^3\right) = 4\pi r^2 \cdot \frac{dr}{dt} = -k\cdot(4\pi r^2)$$

$$\frac{dr}{dt} = -k$$

$$dr = -k\cdot dt$$

$$r(t) = -kt + \underbrace{C}_{=r_0} = 1 - \frac{1}{6}t$$

$r_0 = 1$, 3개월 후 $r = 0.5$

$$0.5 = 1 - 3k$$

$$k = \frac{1}{6} \;//$$

$$0.05 = 1 - \frac{1}{6}t$$

$$\frac{1}{6}t = 0.95$$

$$t = 5.7$$

$\therefore$ $3 + 5.7 = 8.7$ (개월)

06. < 녹지 않는 설탕의 양 : $y(t)$

녹는 속도 : $\frac{dy}{dt}$

$$\frac{dy}{dt} = ky$$

$$\frac{1}{y}dy = k\cdot dt$$

$$\ln|y| = kt + C$$

$$C\cdot e^{kt} = y \;//$$

$$y(0) = \frac{1}{100}$$

$$y(4) = \frac{1}{300}$$

$$y(0) = C = \frac{1}{100}$$

$$y(4) = \frac{1}{100}\cdot e^{4k} = \frac{1}{300}$$

$$\therefore e^{4k} = \frac{1}{3}$$

$$4k = \ln\frac{1}{3}$$

$$k = \frac{1}{4}\ln\frac{1}{3}$$

$$\therefore y(t) = \frac{1}{100}e^{\frac{1}{4}\ln\left(\frac{1}{3}\right)\cdot t}$$

$$y(20) = \frac{1}{100}\cdot e^{\frac{1}{4}\ln\left(\frac{1}{3}\right)\cdot 20}$$

$$= \frac{1}{100}e^{5\ln\frac{1}{3}}$$

//